MOS Device Capacitances:

The basic quadratic I/V relationship derived in the previous section along with corrections for body effect and channel-length modulation provide a reasonable model for understanding the "dc" behavior of CMOS circuits. In many analog circuits, however, the capacitances associated with the devices must also be taken into account so as ~~the~~ to predict the "ac" behavior as well.

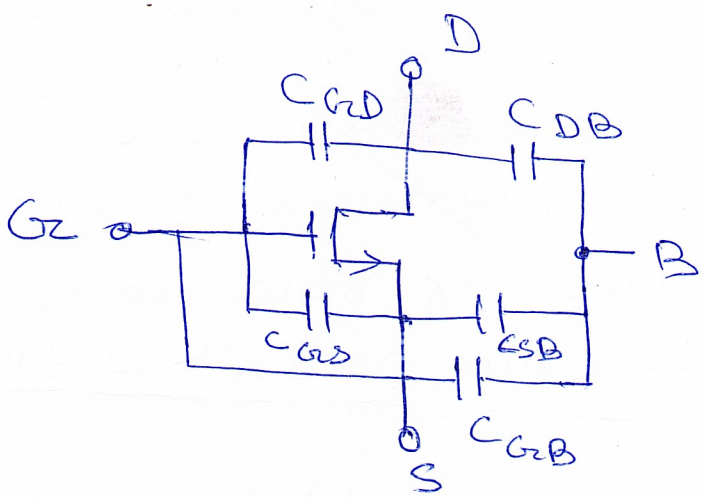


Fig. MOS capacitance.

We expect that a capacitance exists between every two of the four terminals of a MOSFET. Moreover, the value of each of these capacitance may depend on the bias conditions of the transistor. Considering the physical structure in Fig(a) below we identify the following.

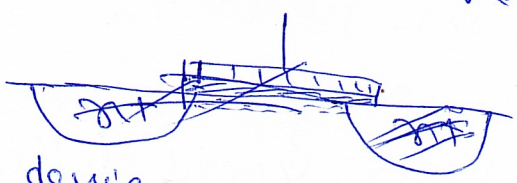


Fig: MOS device capacitances.

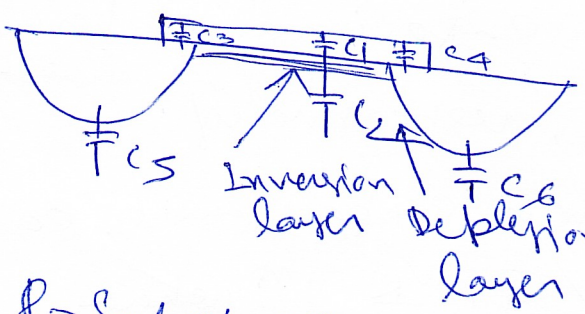
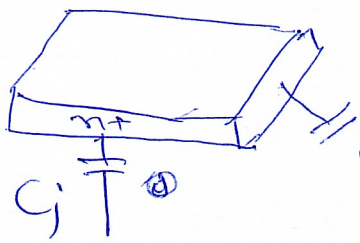


Fig: de composition of S/D junction capacitance into bottom-plate and side wall components.

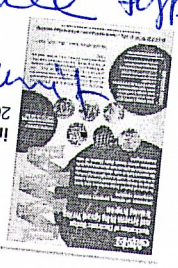


(1) oxide capacitance beth the gate and the channe
$C_1 = WLC_{ox}$;

(2) Depletion capacitance beth the channel and th
Substrate, $C_2 = WL\sqrt{\varepsilon_{si} N_{sub}/(4\phi_F)}$.

(3) Capacitance due to the overlap of the gate wi
with the source and drain areas, C_3 and C_4.
Owing to fringing electric field lines, C_3 and C_4
cannot be simply written as WLC_{ox}, and
are usually obtained by more elaborate calculatio
The overlap capacitance per unit width is denoted
by C_{ov};

(4) Junction capacitance beth the source/drain area
and the substrate. as Fig.(b). This capacitance is
usually decomposed into two components: bottom
-plate capacitance associated with the bottom of
the junction, C_j and sidewall capacitance due to
the perimeter of the junction, C_{jsw}. The
distinction is necessary because different
transistor geometries yield different area and
perimeter values for the S/D junctions. We typically
specify C_j and C_{jsw} as capacitance per unit
area and unit length, respectively.
Note that each junction capacitance can be
expressed as $C_j = C_{j0}/\left[1 + \frac{V_R}{\phi_B}\right]^m$, where



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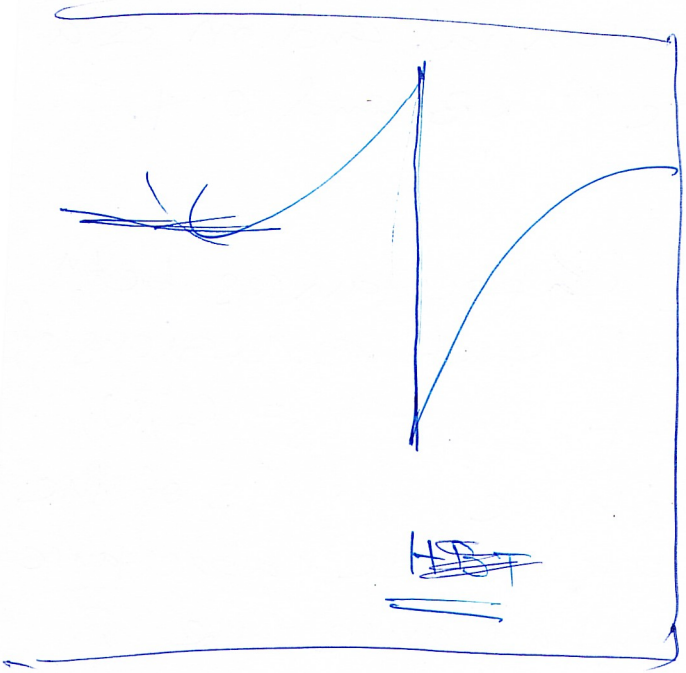
V_R is the reverse voltage across the junction, ϕ_B is the junction built-in-potential, and m is a power typically in the range of 0.3 and 0.4.

We now ~~consider~~ derive the capacitances beth terminals of a MOSFET in different operation regions. If the device is off, $C_{GD} = C_{GS} = C_{ov}W$, and the gate-bulk capacitance consists of the series combination of the gate oxide capacitance and the depletion region capacitance, i.e., $C_{GB} = (WLC_{ox})C_d / (WLC_{ox} + C_d)$, where L is the effective length and $C_d = WL\sqrt{q\epsilon_{Si}N_{sub}/(4\phi_F)}$. The value of C_{SB} and C_{DB} is a function of the source and drain voltages with respect to the substrate.

If the device is deep triode region, i.e. if S and D have approximately equal voltages, then the gate-channel capacitance, WLC_{ox}, is divided equally beth the gate and source terminals and the gate and drain terminals. This is because a change ΔV in the gate voltage draws equal amounts charge from S and D. Thus $C_{GD} = C_{GS} = WLC_{ox}/2 + WC_{ov}$.

If in saturation, a MOSFET exhibits a gate-drain capacitance of roughly WC_{ov}. The potential difference beth the gate and the channel varies from V_{GS} at the source to $V_{GS} - V_{TH}$ at the pinch-off point, resulting in a nonuniform vertical electric field in the gate

oxide along the channel. It can be proved that the equivalent capacitance of the structure excluding the gate-source overlap capacitance is $2WLC_{ox}/3$. Thus $C_{gs} = 2WLC_{ox}/3 + WC_{ov}$.



The behavior of C_{GD} and C_{GS} in different regions of operation is plotted in Fig. below. Note that the above equations do not provide a smooth transition from one region of operation to another, creating convergence difficulties in simulation programs.

The gate-bulk capacitance is usually neglected in the triode and saturation regions because the inversion layer acts as a "shield" between the gate and the bulk. In other words, if the gate voltage varies, the charge is supplied by the source and the drain rather than the bulk.

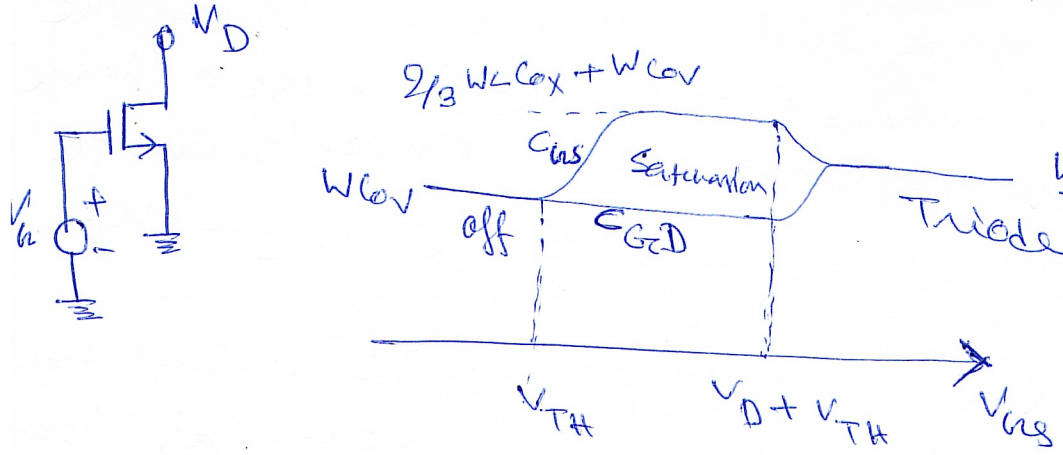


Fig. Variation of gate-source and gate-drain capacitances versus V_{GS}.